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Does Climate and Biodiversity Sentiment Drive Volatility in Financial Markets? Evidence from Green Stocks

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ABSTRACT

We study the predictive role of climate sentiment on the volatility of green equities. Utilizing news-based measures of climate and biodiversity risks and high-frequency ETF data, we document a significant volatility effect of climate sentiment in green equities. While market attention to climate change is more important at shorter forecast horizons, attention to developments regarding biodiversity loss is a stronger predictor of volatility at longer forecast horizons. Economic analysis further shows that climate-focused investors can significantly improve the risk and return tradeoff in their green investment strategies by incorporating climate and biodiversity risk proxies in their volatility forecasting models.

KEYWORDS

Biodiversity risk; Climate risk; Green stocks; MCS test; Volatility forecasting

Introduction

There is growing evidence that climate risk has now emerged to be a significant concern when making investment decisions (Krueger, Sautner, and Starks 2020; Reboredo and Otero 2021; Ceccarelli, Ramelli, and Wagner 2023) as climate risk is shown to impact economic growth prospects (e.g. Stern 2007) and firm profitability (e.g. Addoum, Ng, and Ortiz-Bobea 2020; Pankratz, Bauer, and Derwall 2019). From an asset pricing perspective, a well-established line of works now shows that climate risk exposure translates into a significant risk premium demanded by investors on equity (e.g. Bolton and Kacperczyk 2021; Hsu, Li, and Tsou 2023; Faccini, Matin, and Skiadopoulos 2023), bond (e.g. Painter 2020; Huynh and Xia 2021) and real estate markets (e.g. Baldauf, Garlappi, and Yannelis 2020; Murfin and Spiegel 2020; Bernstein, Gustafson, and Lewis 2019). In this regard, green equities offer investors a means to diversify conventional portfolios with assets that can potentially hedge climate related risks in their investments. The construction of optimal portfolios that incorporate green equities, however, requires accurate modeling and forecasting of volatility in this market segment as volatility is a key input to investment decisions and

portfolio allocations (Poon and Granger 2003). Despite the multitude of studies that examine the nexus between climate risk and financial markets, however, the predictive role of climate risks over volatility dynamics in green equities and whether or not the predictive relationships can be used in optimal asset allocations is largely understudied. This paper extends the emerging literature on climate finance from a novel perspective by exploring the predictive role of climate sentiment on the volatility of green equity returns. By doing so, we provide novel insight into how volatility forecasts in these assets can be improved *via* measures of climate sentiment and whether the predictive relationships can be used to extract economic gains.

By the very nature of their business dynamics, green equities are often exposed to not only the risks associated with the prospects of the new, sustainability focused technologies they invest in, but also the risks associated with the changing regulatory/political environment related to climate regulations. Accordingly, the nature of their industry makes it especially challenging to model price fluctuations in this growing segment of the equity market. Clearly, modeling volatility dynamics in these assets is a critical concern in the optimal asset allocation decisions toward these

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assets, especially considering the growing evidence that sustainability is increasingly viewed as a necessity rather than a luxury good (Albuquerque et al. 2020; Pástor and Vorsatz 2020; Yousaf, Suleman, and Demirer 2022). Furthermore, given that these assets offer a viable alternative to conventional assets regarding their diversification benefits against climate related risks in equity portfolios (Cepni, Demirer, and Rognone 2022), accurate volatility forecasts in these assets are crucial for effective hedging strategies. In that regard, recent studies provide us with several guidelines as to what could be of interest when it comes to improving volatility forecasts in this growing segment of the equity market. For example, Demirer and Prodromou (2025) show that measures of climate uncertainty extracted from textual analysis of articles published in the media can help predict the magnitude and direction of intraday jumps in green assets. Similarly, examining five green equity sectors within a DCC-MIDAS framework, Polat, Demirer, and Ekşi (2024) show that investor sentiment and climate uncertainty play an important role as a driver of systematic risk exposures in green equities. These findings support the earlier evidence on conventional stocks that links investor sentiment to price fluctuations (e.g. Baker and Wurgler 2006; Huang et al. 2015) as well as the recent evidence that relates climate factors to volatility in industry returns (Luo et al. 2025).

Sentiment and uncertainty are distinct yet interconnected concepts, particularly in the context of financial markets and environmental issues. Sentiment refers to the general emotional disposition or attitude toward a particular subject, such as climate change or biodiversity loss, which is often conveyed through the tone of media coverage, categorized as positive, negative, or neutral. It is commonly quantified using natural language processing techniques, such as the Bidirectional Encoder Representations from Transformers (BERT) model, which aims to capture the public sentiment embedded in news articles (Giglio et al. 2023). In contrast, uncertainty is associated with the lack of predictability regarding future events or outcomes, frequently associated with risk. This is typically measured using indices like the volatility index (VIX), which reflects market expectations of future volatility (Coqueret, Giroux, and Zerbib 2025). While sentiment can influence investor behavior by shaping their confidence in specific assets, uncertainty is more likely to induce market volatility and risk aversion, as investors demand higher returns to compensate for increased risk. In the realm of

climate and biodiversity news, sentiment gauges public concern, whereas uncertainty encapsulates the unpredictability of environmental developments and their potential economic consequences. In our application, climate sentiment is gauged using a climate risk index, which reflects the level of awareness and concern regarding climate change issues. Similarly, biodiversity sentiment is assessed through a biodiversity attention index, which measures the level of attention and concern directed toward biodiversity loss.¹

Previous studies that are related to our context have largely focused on the physical and transitional risks posed by climate change, as exemplified in works such as those by Engle et al. (2020), Cepni et al. (2023) and Bua et al. (2024), among others. However, in a recent study, Giglio et al. (2023) identify biodiversity risk as an important new dimension of risk that is distinct from climate risk, associating biodiversity risk with the threats to the variety of life on earth that can impact the production of food and nature-based materials as well as the production of medicines and drugs obtained from natural compounds. In contrast, although related to some extent, climate risk relates more to the negative effect of a change in climate patterns which can have distinctly different effects on businesses compared to biodiversity related risks. Indeed, constructing news-based measures of biodiversity and climate risks by grouping articles published in the New York times into positive, negative or neutral sentiment categories, Giglio et al. (2023) document only weak correlations between climate and biodiversity risk exposures across industries in the cross-section, while their analysis also reveals that portfolios designed to hedge against climate risk do not effectively mitigate biodiversity risk, and vice versa. Accordingly, this underscores the critical importance of distinguishing between biodiversity and climate risks in economic models, although there has been relatively little exploration into the effects of biodiversity risks on asset valuations and economic activities.

Our analysis focuses on green equities and ETFs, which are specifically constructed to include firms with material exposure to environmental risks and opportunities, including those related to biodiversity. The distinction between climate and biodiversity related risks requires separate strategies for risk assessment and management, particularly for investors in green equities. For these firms, biodiversity-related regulation, public attention, and supply chain dependencies can have tangible financial impacts. Even for sectors with less direct exposure, shifts in market

sentiment about biodiversity can influence investor flows, risk premia, and sectoral allocations. Giglio et al. (2023) find that biodiversity risks already influence asset values, with portfolios sorted on firms' biodiversity risk exposures showing a positive relationship with changes in aggregate biodiversity risk, indicating that these risks are being recognized in the market. These risks also include both physical risks, such as the degradation of ecosystem services, and regulatory risks, like stricter land-use regulations, which can directly impact industries involved in green equities, such as renewable energy. According to Giglio et al. (2023), the perception among market participants is that biodiversity risks are not yet adequately priced in equity markets, suggesting potential future market corrections as awareness and understanding of these risks increase. Although biodiversity and climate risks are distinct, they are interconnected. Biodiversity loss can exacerbate climate change by reducing carbon sinks, while climate change can further threaten biodiversity. This interconnection underscores the need for comprehensive risk management strategies in green equities. For green equity investments, which are often seen as part of the solution to environmental challenges, understanding and addressing biodiversity risk is essential.

This paper contributes to the literature in climate finance in novel aspects by examining (i) the predictive role of sentiment regarding climate change and biodiversity loss over volatility dynamics in green equities; and (ii) whether such a predictive relation can be used to generate economic results *via* a forward-looking investment strategy. This is clearly an important consideration as recent evidence suggests that green equities can help diversify conventional stock portfolios in a meaningful way, particularly against climate related exposures (Cepni et al. 2023; Polat, Demirer, and Ekşi 2024). Furthermore, distinguishing between climate and biodiversity related risks is particularly important in the context of green equities as green firms, particularly those that focus on renewable energy, would be more affected by biodiversity regulation than climate regulation as the operations of these firms would have a more direct impact on ecosystems and habitats rather than climate change. At this point, it must be noted that even though firms individually do not have power to impact policy driven uncertainty as this is completely out of their realm, investors can still assess risk exposures with respect to this uncertainty and design hedge portfolios to mitigate the negative impact of this type of uncertainty in their investment portfolios.

The key here, however, is to assess these risk exposures and design investment strategies that will mitigate the negative impact of these risks in investment returns and that is our focus in this paper. Even though firms individually do not have the power to impact climate related developments, investors can assess the impact of those risks on financial market returns and design risk management strategies to mitigate those risks in their portfolios.

As noted by Giglio et al. (2023), renewable energy firms, particularly in solar and wind, are susceptible to biodiversity regulations due to their land use, which can disrupt local ecosystems. Regulations focusing on habitat and species protection often impose stricter requirements than climate regulations, which primarily address emissions. Furthermore, the business operations of some green equity sectors may be more exposed to biodiversity risk than risks associated with climate change, as their operations might be dependent on particular ecosystems or species that are not necessarily affected by climate change. In that regard, biodiversity risks are distinct from climate risks, with the former directly affecting ecosystems, require more immediate operational adjustments, which could be particularly disruptive for renewable energy firms. Accordingly, these arguments justify separately analyzing the predictive role of climate and biodiversity related risks in price fluctuations observed in green equity markets. To our knowledge, ours is the first study to examine the predictive impact of both biodiversity risk and climate risks on realized volatility, thereby elucidating their respective roles in this context.

Utilizing two alternative measures of climate sentiment, i.e. the NYT climate and biodiversity news indexes, recently proposed by Giglio et al. (2023) based on a textual classification of positive and negative news regarding climate and biodiversity concerns, and high frequency data on green ETFs within the heterogeneous autoregressive (HAR) framework of Corsi (2008), our findings show that climate and biodiversity related sentiment have significant predictive power over volatility in green equity returns although these proxies capture markedly different aspects of sentiment related to climate concerns. We find that climate sentiment that relates to risks associated with the potential negative consequences of a change in the climate system has an immediate impact on short-term volatility in green equities. In contrast, our findings reveal that sentiment related to biodiversity loss associated with the threats to the variety of life and the ecosystem, has a significant impact on longer-term

volatility in green equity returns. From the perspective of the efficient market hypothesis, our findings indicate that investors respond quickly to climate risks, while they gradually incorporate biodiversity-related factors into their trades of green stocks.

While the predictive power of climate related sentiment over volatility extends to the out-of-sample volatility forecasts, we also find that the observed predictive relationship can be used to extract economic benefits, particularly, when it comes to bad volatility forecasts in these assets. This is an important consideration as investors can utilize our findings to improve downside risk forecasts at short forecast horizons. Our results show that volatility models that include climate sentiment offer optimal performance for short and intermediate-term forecasts. Additionally, models integrating both climate and biodiversity sentiment highlight the benefits of sustainable investment strategies in the long term, emphasizing the importance of diversified model selection and risk management. Finally, exploring the impact of climate and biodiversity sentiment on good and bad volatility in green equities across different time frames, we show that climate sentiment has consistent predictive power across all forecast horizons, while biodiversity sentiment forecasts intermediate and long-term volatility across the directional volatility models. These findings underscore the importance of environmental factors in financial decision-making and advocate for effective regulatory oversight to manage sentiment-driven volatility and enhance transparency in environmental data disclosure.

Considering the growing importance of green equities in investment decisions, our findings emphasize the importance of environmental factors in financial decisions, necessitating regulatory oversight. While sentiment is found to have an impact on both good and bad volatility forecasts, our results suggest the potential amplification of volatility in green equities by positive developments in the green energy sector, contingent on market dynamics and sentiment. Furthermore, the economic significance of our findings across forecast horizons emphasizes the effectiveness of models that incorporate climate and biodiversity factors in optimizing investment performance. Drawing parallels with Baur (2012), our findings highlight the potential amplification of volatility in green equities by positive developments in the green energy sector. Finally, evaluating the economic significance of good and bad volatility across the short and long forecast horizons, our analysis shows that models focusing on good volatility yields higher risk adjusted returns for short and intermediate-term forecasts,

while models incorporating climate and biodiversity factors outperform others for longer forecast horizons, emphasizing their efficacy in capturing market dynamics and optimizing investment performance.

The remainder of the paper is structured as follows. Section “Data and methodology” describes the methodology and data used in our empirical analysis. Section “Empirical findings” presents the empirical findings from the volatility prediction models and the economic implications of predictability. Section “Extension to directional volatility” provides evidence from asymmetric volatility measures and robustness checks. Finally, Section “Conclusion” concludes with directions for future research.

Data and methodology

Data

We analyze price data at a 5-min frequency from November 2009 to June 2022, sourced from the Thompson Reuters Tick History database (TRTH). To capture equity market dynamics in green equities, we focus on the Invesco Global Clean Energy ETF (PBD) that is popularly used in the literature as a proxy for green stocks (e.g. Bouri, Iqbal, and Klein 2022). The Invesco Global Clean Energy ETF is designed to mirror the WilderHill New Energy Global Innovation Index, which includes companies engaged in the development of cleaner energy and conservation technologies. The fund allocates a minimum of 90% of its total assets to securities that comprise the index, including American Depositary Receipts (ADRs) and global depositary receipts (GDRs).² The holdings of this ETF are diversified across multiple sectors, including renewable energy, energy efficiency, and advanced materials, thus offering diversified exposure to a wide array of clean energy sectors, including wind, solar, hydro, and advanced materials.³

Daily returns are computed as the difference in closing prices (New York time) and intraday returns are obtained using last-tick interpolation based on 1-min prices (if the price is not available at the 1-min stamp, the previously available price is imputed). Next, we compute 5-min returns as the log differences of these prices, which in turn are used to generate the realized series. After applying several filters including the removal of shortened trading days, extreme jumps and matching the dates with the climate and biodiversity index, our final sample yields 3,126 daily observations. The in and out of sample analysis is conducted by dividing the whole sample into two sub-samples for in-sample estimation (November 4, 2009 –

November 21, 2018) and out-of-sample prediction (November 22, 2018 – June 30, 2022).

As mentioned earlier, our forecasting models utilize the NYT Biodiversity news and NYT Climate news indexes recently proposed by Giglio et al. (2023) as a proxy for public sentiment regarding climate concerns.⁴ These indexes are constructed based on a corpus of daily news articles published in the New York Times. The construction of the biodiversity news index begins with compiling a biodiversity dictionary containing terms such as ecosystem, habitat, species, and deforestation. These terms are selected based on their similarity to “biodiversity” using Google’s word2vec implementation. Utilizing this dictionary, sentences are tagged as biodiversity-related if they contain at least one of these terms, excluding irrelevant combinations like “software ecosystem.” Articles containing a minimum of two biodiversity sentences are then identified as covering biodiversity. Considering that biodiversity news can convey either positive or negative sentiments, to differentiate between sentiments, the Bidirectional Encoder Representations from Transformers (BERT) model of Devlin et al. (2019) is employed to classify each selected biodiversity sentence as positive, negative, or neutral. Sentences with positive (negative) sentiment are assigned a score of +1 (−1) and neutral sentences are marked with 0. Articles are then categorized as having positive or negative sentiment based on the average sentiment scores of all biodiversity sentences within them. This process yields approximately 6.6% of articles classified as positive, 77.9% as neutral, and 15.7% as negative. Subsequently, the NYT Biodiversity news index is calculated as the number of negative biodiversity articles minus the number of positive ones on any given day, with higher values indicating a greater prevalence of negative news about biodiversity risks.

Similarly, Giglio et al. (2023) construct a climate news series, namely the NYT Climate news index, by applying the same methodology to climate-related sentences that include terms such as “climate change” and “global warming”. The authors note that the climate and biodiversity news indexes capture distinctly different aspects of public sentiment related to climate concerns with a correlation of only 22% between the two indexes. Although both series are generated with the same method and the same corpus of news articles, they observe that periods with negative biodiversity-related news do not necessarily correspond to periods with negative climate news. As an example, news related to Bush’s decision to withdraw from the Kyoto Protocol in March of 2001 or Trump’s announcement in June 2017 to withdraw from the Paris Climate Agreement did not

result in spikes in the biodiversity index whereas the same news resulted in local spikes in the time series of the NYT climate index. The authors also note that the biodiversity news index has a low correlation with the climate risk indexes recently proposed by Faccini, Matin, and Skiadopoulos (2023), obtained *via* textual and narrative analysis of Reuters climate-change news, in the range of −0.2 to 0.18. Accordingly, these observations justify separately analyzing the predictive information captured by the climate and biodiversity sentiment among green ETF traders over subsequent volatility patterns.

At this point, it is important to note that while the underlying physical processes of climate change and biodiversity loss are gradual and manifest over long horizons, our study focuses on the market sentiment and investor attention to these risks, as proxied by news-based indices. This setting aligns well with the forward-looking nature of financial markets in that investors react not only to slow-moving fundamentals but also to the arrival and interpretation of new information. News about climate or biodiversity, such as regulatory changes, policy announcements, or major reports, can trigger immediate reassessments of risk, even if the underlying physical process is slow. This is in fact, consistent with the literature on news shocks and market reactions in financial markets (e.g., Engle et al. 2020; Giglio et al. 2023) and supports the well-documented evidence on the impact of investor sentiment on financial markets (e.g. Baker and Wurgler 2006; Huang et al. 2015) with recent evidence showing that news-based climate sentiment indices can predict short-term jumps and volatility in green asset prices (e.g., Demirer and Prodromou 2025; Polat, Demirer, and Ekşi 2024). Our focus also aligns with the recent finding in Giglio et al. (2023) that biodiversity risks are not yet adequately priced in equity markets, suggesting potential future market corrections as awareness and understanding of these risks increase. In this regard, our approach builds on the literature showing that sentiment acts as a channel through which slow-moving risks are rapidly priced in financial markets.

Furthermore, it is important to critically assess the informational content of the sentiment indices and their relevance for asset pricing and volatility forecasting. The sentiment indices employed in this study (NYT Climate and Biodiversity News Indices) are constructed from a large corpus of news articles, systematically classified using advanced natural language processing (NLP) techniques. These indices capture the aggregate tone and media attention regarding climate and biodiversity risks, reflecting the information environment faced by investors. While one might

question whether sentiment merely represents “noise” or is truly related to fundamental risk, there are several reasons to expect that sentiment can influence returns and volatility. First, financial markets react not only to slow-moving fundamentals but also to the arrival and interpretation of new information. News and sentiment can act as signals about future risks, regulatory changes, or shifts in public and investor attention, prompting immediate market reactions. Second, if investors perceive an increase in climate or biodiversity risk (as reflected in negative sentiment), they may demand a higher risk premium, adjust their portfolios, or hedge exposures, leading to changes in prices and volatility, even in the short term.

Finally, it must be noted that while climate and biodiversity risks are fundamentally long-term in nature, our use of high-frequency data is motivated by the need to accurately measure realized volatility and to capture the immediate market response to new information. Financial markets react in real time to news and changes in sentiment, even about long-horizon risks. High-frequency data allows us to detect short-term volatility spikes and jumps that may follow the release of climate- or biodiversity-related news, as established in the volatility forecasting literature (e.g., Andersen and Bollerslev 1998; Corsi 2008). Although not all investors act immediately on every piece of climate news, the presence of market participants, such as ESG funds, climate-focused investors, or risk managers, who are sensitive to such information, is sufficient to generate observable effects in high-frequency volatility. Our use of high-frequency data is a methodological choice that enhances the precision of volatility measurement and better isolates the impact of news and sentiment shocks. This approach does not imply that climate risk itself is high frequency, but rather that market reactions to information about climate risk can be immediate and measurable, even if the underlying risk is long-term.

Figures 1 and 2 present the time series plots for various realized volatility variations along with the climate and biodiversity sentiment indices. A notable spike in all volatility series is observed around the 2008 financial crisis period, suggesting that green equities were also significantly affected by this market shock. Figure 2 shows the time series of climate and biodiversity indices used in our volatility forecasting models. Note that these indexes are calculated as the number of negative articles minus the number of positive ones on any given day, with higher values indicating a greater prevalence of negative news about biodiversity (or climate) risks. Table 1 shows the summary statistics for the various realized volatility measures obtained from intraday data and the climate and

biodiversity news indices, denoted by CLMT and BIO, respectively. In general, the realized volatility series exhibit right-skewness and fat tails along with significant dynamic dependencies in daily, weekly, and monthly components, implied by the positive Ljung-Box Q-statistics statistics. Separately, we observe larger values for the average biodiversity index values compared to the climate index, implying the prevalence of negative news about biodiversity concerns.

Realized volatility predictors

As the main focus of our forecasting models, realized variance (RV) is obtained as the sum of squares of intraday returns as per Andersen and Bollerslev (1998) using

$$RV_t = \sum_{j=1}^M r_{t,j}^2, \quad (1)$$

where $r_{t,j}$ is the j^{th} intraday return ($j = 1, 2, \dots, M$) on day t and $1/M$ is the sampling frequency. Given the established role of jumps as a volatility predictor, we then compute the discontinuous jump (J_t) following Andersen, Dobrev, and Schaumburg (2012), formulated as

$$J_t = RV_t - MedRV_t, \quad (2)$$

where $MedRV_t$ is the median realized variance as this measure possesses better efficiency properties and robustness to jumps and small returns, formulated as:

$$MedRV_t = \frac{\pi}{6 - 4\sqrt{2} + \pi} \left(\frac{M}{M-2} \right) \sum_{j=2}^{M-1} med(|r_{t-1+(j-1)/M}|, |r_{t-1+j/M}|, |r_{t-1+(j+1)/M}|)^2, \quad (3)$$

where med denotes the median function. To detect jumps that are statistically significant, we use the jump test statistic Z_t^{med} of Andersen, Dobrev, and Schaumburg (2012), formulated as:

$$Z_t^{med} = \frac{(RV_t - MedRV_t)/RV_t}{\sqrt{\left(\left(\frac{\pi}{2}\right)^2 + \pi - 5\right) \frac{1}{M} \max\left(1, \frac{MedRQ_t}{MedRV_t}\right)}} \rightarrow N(0, 1), \quad (4)$$

where $MedRQ_t$ is the median realized quarticity computed as:

$$MedRQ_t = \frac{3\pi M}{9\pi + 72 - 52\sqrt{3}} \left(\frac{M}{M-2} \right) \sum_{j=2}^{M-1} med(|r_{t-1+(j-1)/M}|, |r_{t-1+j/M}|, |r_{t-1+(j+1)/M}|)^4. \quad (5)$$

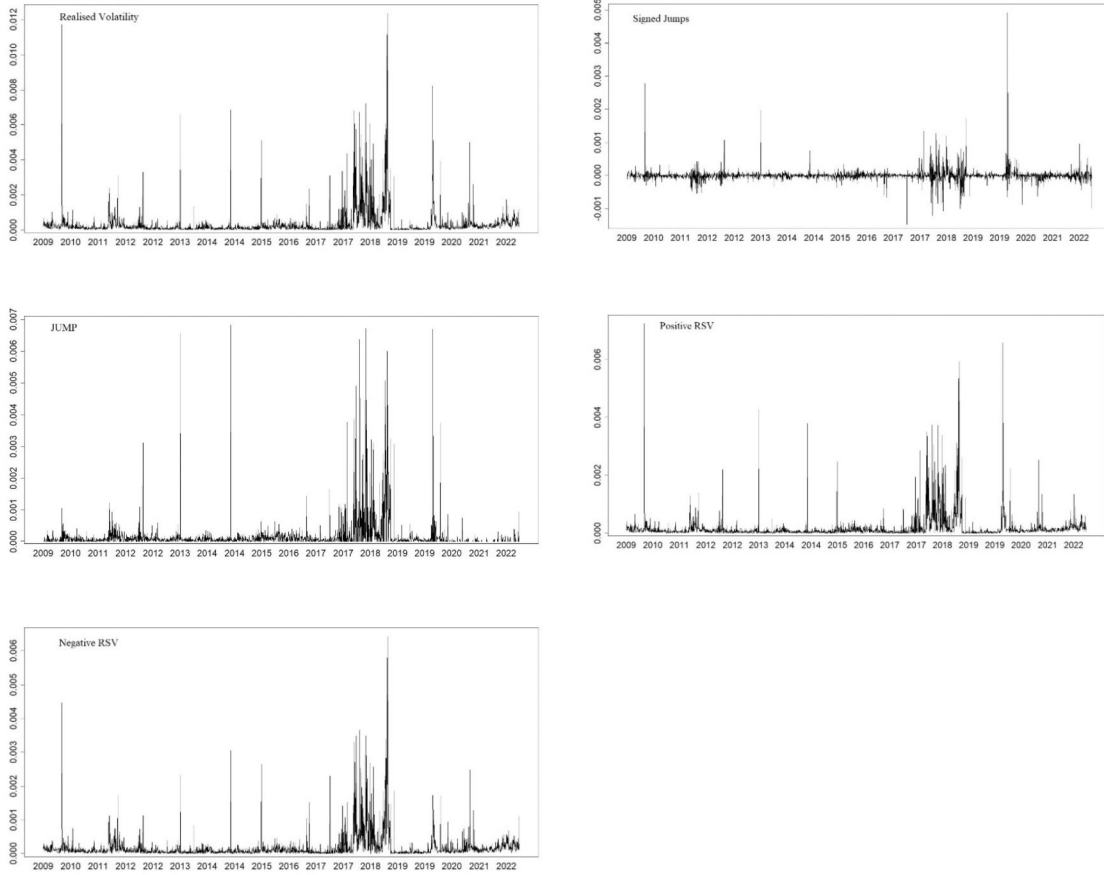


Figure 1. Time-series of realized volatility components obtained from intraday returns.

Note: Figure 1 presents the plots for the daily realized volatility and jump measures computed using intraday data for the Invesco Global Clean Energy ETF as described in Section “Data”.

Once the realized jump series are obtained, we generate the statistically significant jumps as

$$J_t = I(Z_t^{med} > \Phi_\alpha) \cdot (RV_t - MedRV_t), \quad (6)$$

where $I(\cdot)$ denotes the indicator function with the trigger value (Φ_α) at the significance level of α ($\alpha = 0.999$) is based on the standard normal distribution. Building on Barndorff-Nielsen, Kinnebrock, and Shephard (2008) and Patton and Shephard (2015), we also include the realized semi-variance measures as predictors to improve the performance of volatility prediction models and so compute the positive and negative realized semi-variance series as $RSV_t^+ = \sum_{j=1}^M I\{r_{t,j} > 0\} r_{t,j}^2$ and $RSV_t^- = \sum_{j=1}^M I\{r_{t,j} < 0\} r_{t,j}^2$, respectively. Finally, to improve asymmetric volatility forecasts, following Patton and Shephard (2015), daily signed jump (SJ_t) is computed as the difference between the positive and negative realized semi-variances, i.e. $SJ_t = RSV_t^+ - RSV_t^-$, which are then utilized as an additional predictor in the volatility models.

Volatility forecasting models

Benchmark forecasting models

The benchmark forecasting model follows the heterogeneous autoregressive (HAR) model proposed by Corsi (2008) that is widely used in the context of realized volatility forecasting, particularly in high frequency applications. In this study, we focus on the predictive role of climate sentiment proxies over volatility in green equities and so build our climate-based model on the benchmark models described in this section. Following the evidence in Niu et al. (2024) of a greater effect of negative returns on future volatility than positive returns, we also incorporate in the benchmark model a leverage term, leading to the LHAR-RV specification as the benchmark forecasting model, formulated as

$$\begin{aligned} \overline{RV}_{t+H} = & \alpha_0 + \alpha_1 RV_t + \alpha_2 RV_{t-5,t} + \alpha_3 RV_{t-22,t} + \phi_1 r_t^{d-} \\ & + \phi_2 r_t^{w-} + \phi_3 r_t^{m-} + \varepsilon_{t+H} \end{aligned} \quad (7)$$

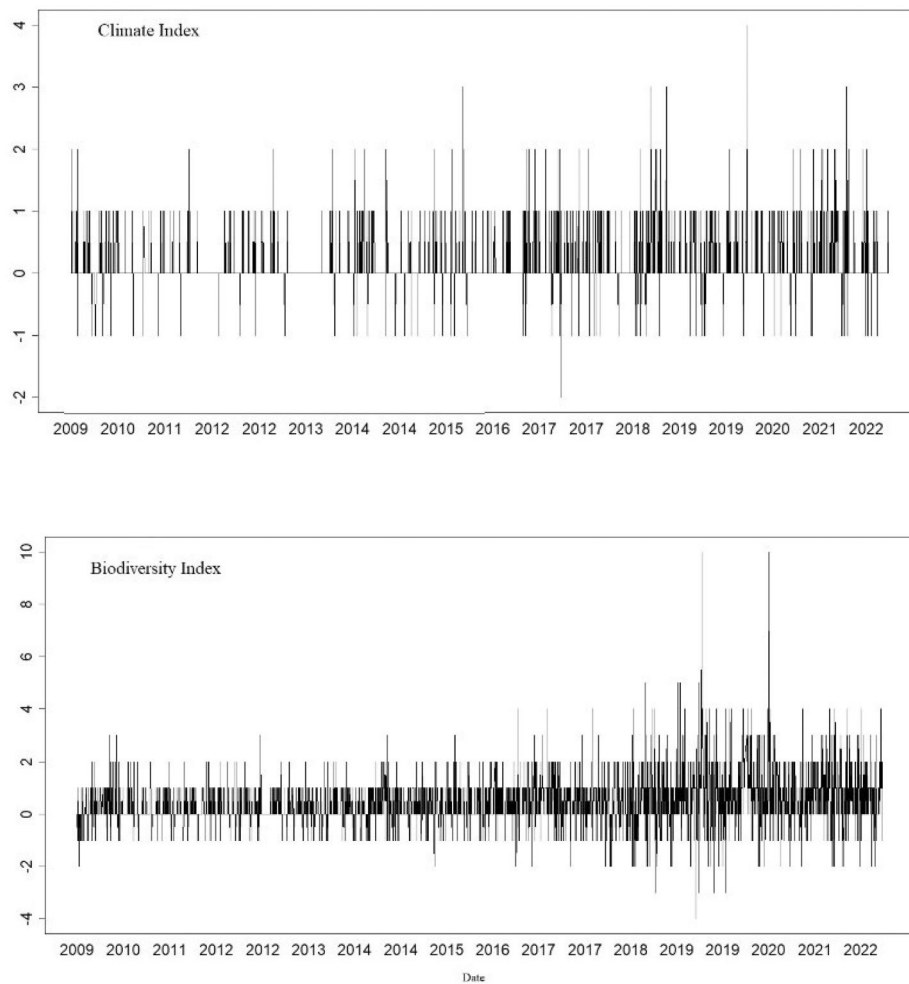


Figure 2. Time-series of climate and biodiversity index.

Note: Figure 2 shows the daily New York Times Climate and Biodiversity news indices from Giglio et al. (2023). Each index is computed by subtracting the number of negative articles from the number of positive articles on a given day, so higher values reflect a stronger presence of negative news about climate or biodiversity risks.

Table 1. Summary statistics of volatility components and the climate and biodiversity indexes.

	Mean	Median	Std. Dev	Min	Max	Skewness	Kurtosis	Q (10)	Q (20)
RV_t	0.03599	0.01597	0.07817	0.00019	1.23498	6.98703	67.54270	4990***	7547***
RV_{t-5t}	0.03597	0.01593	0.07823	0.00019	1.23498	6.98268	67.44540	4983***	7537***
RV_{t-22t}	0.03601	0.01580	0.07844	0.00019	1.23498	6.96353	67.06600	4956***	7496***
J_t	0.01594	0.00540	0.04966	0.00000	0.68336	7.77452	75.56550	2121***	3675***
RSV_t^+	0.01801	0.00756	0.04200	0.00003	0.72404	7.68737	83.26380	3903***	5981***
RSV_{t-5t}^+	0.01800	0.00753	0.04203	0.00003	0.72404	7.68281	83.14900	3897***	5974***
RSV_{t-22t}^+	0.01802	0.00751	0.04214	0.00003	0.72404	7.66211	82.68880	3877***	5941***
RSV_t^-	0.01798	0.00788	0.03816	0.00004	0.64292	6.69553	63.49850	5336***	7948***
RSV_{t-5t}^-	0.01797	0.00786	0.03819	0.00004	0.64292	6.69128	63.40500	5329***	7937***
RSV_{t-22t}^-	0.01799	0.00781	0.03829	0.00004	0.64292	6.67298	63.04930	5300***	7894***
SJ_t	0.00003	-0.00008	0.01815	-0.14808	0.49226	8.27069	205.10200	55.26***	65.37***
r_t^{d-}	-0.00702	0.00000	0.01369	-0.26313	0.00000	-4.96657	54.85620	511.5***	672.2***
r_t^{w-}	-0.00701	-0.00509	0.00776	-0.11287	0.00000	-4.10312	35.63410	8053***	9361***
r_t^{m-}	-0.00699	-0.00569	0.00522	-0.04491	0.00000	-2.59829	11.73900	24479***	35252***
$CLMT_t$	0.10784	0.00000	0.42439	-2.00000	4.00000	2.29242	10.63440	57.48***	74.64***
$CLMT_{t-5t}$	0.10794	0.00000	0.20935	-0.60000	1.40000	1.68441	4.68781	4964***	5065***
$CLMT_{t-22t}$	0.10754	0.09091	0.11678	-0.09091	0.72727	1.49167	2.82187	21194***	27652***
BIO_t	0.40992	0.00000	0.99870	-4.00000	10.00000	1.51066	8.29497	329.8***	497.4***
BIO_{t-5t}	0.41044	0.40000	0.56041	-1.00000	6.20000	2.44143	15.43870	6888***	8310***
BIO_{t-22t}	0.408842	0.318182	0.358878	-0.22727	2.454545	1.836434	5.309335	23979***	35836***

Note: Table 1 reports summary statistics for the intraday-based measures of realized volatility (RV), jumps (J), and their signed components (RSV and SJ) for the Invesco Global Clean Energy ETF. CLMT and BIO refer to the New York Times climate and biodiversity news indices, each shown at daily, weekly, and monthly frequencies. (*, **, ***) mark significance at the 10%, 5%, and 1% levels, respectively.

where $\overline{RV}_{t+H} = (RV_{t+1} + RV_{t+2} + \dots + RV_{t+H})/H$, ($H = 1, 5, 22$) represents the 1-day, 1-week and 1-month ahead realized volatility, respectively; RV_t is the daily RV, as defined in Equation (1) and $RV_{t-i,t} = (RV_t + RV_{t-1} + \dots + RV_{t-(i-1)})/i$, ($i = 5, 22$) is the weekly and monthly RV. In this setting, $r_t^{d-} = \min(r_t^d, 0)$ captures the leverage effect with r_t^d denoting the daily return, while r_t^{w-} and r_t^{m-} denote the weekly and monthly negative returns formulated as $r_t^{w-} = \min((r_t + r_{t-1} + \dots + r_{t-4})/5, 0)$, $r_t^{m-} = \min((r_t + r_{t-1} + \dots + r_{t-21})/22, 0)$, respectively.

Following Andersen, Bollerslev, and Diebold (2007), Corsi and Renò (2012) and Niu et al. (2024), the model in Equation (7) is further augmented by incorporating a jump component leading to the extended LHAR-RV-J benchmark model as

$$\begin{aligned} \overline{RV}_{t+H} = & \alpha_0 + \alpha_1 RV_t + \alpha_2 RV_{t-5,t} + \alpha_3 RV_{t-22,t} + \beta_1 J_t \\ & + \phi_1 r_t^{d-} + \phi_2 r_t^{w-} + \phi_3 r_t^{m-} + \varepsilon_{t+H} \end{aligned} \quad (8)$$

where J_t is the discontinuous jump formulated in Equation (6). Finally, following Patton and Sheppard (2015), we estimate a second variation to the benchmark model by incorporating a signed jump predictor, leading to the LHAR-RV-SJ model, formulated as

$$\begin{aligned} \overline{RV}_{t+H} = & \alpha_0 + \alpha_1 RV_t + \alpha_2 RV_{t-5,t} + \alpha_3 RV_{t-22,t} + \beta_1 SJ_t \\ & + \phi_1 r_t^{d-} + \phi_2 r_t^{w-} + \phi_3 r_t^{m-} + \varepsilon_{t+H}, \end{aligned} \quad (9)$$

where SJ_t is the daily signed jump obtained by subtracting the negative realized semi-variance from the positive component. Overall, the process yields three benchmark models used to test the predictive performance of climate sentiment based predictive models against the standard HAR-type forecasting models employed in the literature.

Sentiment based (LHAR-type-CLMT and LHAR-type-BIO) forecasting model

In our analysis, we are especially interested in the predictive value of climate sentiment over volatility in green equity markets and so use two alternative proxies that capture public attention to climate related concerns, namely the NYT Biodiversity News Index and NYT Climate News Index, constructed based on textual analysis of news articles. These indexes have been shown in recent studies to have a significant effect in the cross-section of equity returns (Giglio et al. 2023), while commanding a significant risk premium embedded in equity valuations (Garel et al. 2024). In a recent study, Giglio et al. (2023) construct the NYT biodiversity news index

to gauge public sentiment around biodiversity news based on a dictionary containing terms related to biodiversity by grouping articles reported in the New York times into positive, negative or neutral sentiment categories. The biodiversity news index is then calculated as the difference between the number of negative biodiversity articles and the number of positive ones on a specific day such that higher index values indicate more negative news about biodiversity risks. Similarly, they also construct a climate news series, i.e. the NYT Climate News Index, by tracing climate related statements with the terms “climate change” and “global warming”. In our application, we propose several variations to the benchmark models explained in the previous section by incorporating the daily news indexes as an additional predictor and compare the performance of the augmented climate- (CLMT) and biodiversity- (BIO) based against the benchmark.

Giglio et al. (2023) show that portfolios designed to hedge against climate risk do not effectively mitigate biodiversity risk, and vice versa. This highlights the importance of differentiating between biodiversity and climate risks and understanding their individual and combined impacts on predicting realized volatility. Accordingly, we construct three sentiment-based volatility models for each LHAR-type-CLMT and LHAR-type-BIO variation, formulated as

$$\begin{aligned} \overline{RV}_{t+H} = & \alpha_0 + \alpha_1 RV_t + \alpha_2 RV_{t-5,t} + \alpha_3 RV_{t-22,t} + \phi' R_t \\ & + \lambda' Clmt + \varepsilon_{t+H} \end{aligned} \quad (10a)$$

$$\begin{aligned} \overline{RV}_{t+H} = & \alpha_0 + \alpha_1 RV_t + \alpha_2 RV_{t-5,t} + \alpha_3 RV_{t-22,t} + \beta_1 J_t \\ & + \phi' R_t + \lambda' Clmt_t + \varepsilon_{t+H} \end{aligned} \quad (10b)$$

$$\begin{aligned} \overline{RV}_{t+H} = & \alpha_0 + \alpha_1 RV_t + \alpha_2 RV_{t-5,t} + \alpha_3 RV_{t-22,t} + \beta_1 SJ_t \\ & + \phi' R_t + \lambda' SClmt_t + \varepsilon_{t+H} \end{aligned} \quad (10c)$$

where $R_t = [r_t^{d-}, r_t^{w-}, r_t^{m-}]$ and $Clmt_t = [Clmt_d, Clmt_w, Clmt_m]$ capture the daily, weekly and monthly leverage effects and the climate news index, respectively; $\phi_t = [\phi_1, \phi_2, \phi_3]$ and $\lambda_t = [\lambda_1, \lambda_2, \lambda_3]$ are the corresponding coefficients for R_t and $Clmt_t$. In this setting, the predictive relationships at the short-, intermediate- and long-horizons are accounted for by incorporating the weekly and monthly averages of the biodiversity and climate news index, captured by $Clmt_w = (Clmt_t, Clmt_{t-1}, \dots, Clmt_{t-4})/5$ and $Clmt_m = (Clmt_t, Clmt_{t-1}, \dots, Clmt_{t-21})/22$, respectively, while $Clmt_d$ is the daily climate news index.

Augmenting the benchmark models with the NYT biodiversity news index yields three alternative variations of the LHAR-type-BIO models formulated as

$$\begin{aligned} \overline{RV}_{t+H} = & \alpha_0 + \alpha_1 RV_t + \alpha_2 RV_{t-5,t} + \alpha_3 RV_{t-22,t} + \phi' R_t \\ & + \delta' Bio + \varepsilon_{t+H}, \end{aligned} \quad (10d)$$

$$\begin{aligned} \overline{RV}_{t+H} = & \alpha_0 + \alpha_1 RV_t + \alpha_2 RV_{t-5,t} + \alpha_3 RV_{t-22,t} + \beta_1 J_t \\ & + \phi' R_t + \eta Bio_t + \varepsilon_{t+H}, \end{aligned} \quad (10d)$$

$$\begin{aligned} \overline{RV}_{t+H} = & \alpha_0 + \alpha_1 RV_t + \alpha_2 RV_{t-5,t} + \alpha_3 RV_{t-22,t} + \beta_1 SJ_t \\ & + \phi' R_t + \eta' SBio_t + \varepsilon_{t+H}, \end{aligned} \quad (10e)$$

where $Bio_t = [Bio_d, Bio_w, Bio_m]$ represent the daily, weekly and monthly biodiversity news index series, and $\lambda_t = [\lambda_1, \lambda_2, \lambda_3]$ are the corresponding coefficients for R_t and Bio_t . Bio_d is the daily biodiversity index and $Bio_w = (Bio_t, Bio_{t-1}, + \dots + Bio_{t-4})/5$ and $Bio_m = (Bio_t, Bio_{t-1}, + \dots + Bio_{t-21})/22$ are the weekly and monthly biodiversity index, respectively.

Finally, as the most comprehensive forecasting model, we expand the above models by incorporating both the climate and biodiversity indexes as predictors and estimate three variations of the combined LHAR-type-Clmt-Bio forecasting models formulated as

$$\begin{aligned} \overline{RV}_{t+H} = & \alpha_0 + \alpha_1 RV_t + \alpha_2 RV_{t-5,t} + \alpha_3 RV_{t-22,t} + \phi' R_t \\ & + \lambda' Clmt_t + \eta' Bio_t + \varepsilon_{t+H} \end{aligned} \quad (11a)$$

$$\begin{aligned} \overline{RV}_{t+H} = & \alpha_0 + \alpha_1 RV_t + \alpha_2 RV_{t-5,t} + \alpha_3 RV_{t-22,t} + \beta_1 J_t \\ & + \phi' R_t + \lambda' Clmt_t + \eta' Bio_t + \varepsilon_{t+H} \end{aligned} \quad (11b)$$

$$\begin{aligned} \overline{RV}_{t+H} = & \alpha_0 + \alpha_1 RV_t + \alpha_2 RV_{t-5,t} + \alpha_3 RV_{t-22,t} + \beta_1 SJ_t \\ & + \phi' R_t + \lambda' Clmt_t + \eta' Bio + \varepsilon_{t+H} \end{aligned} \quad (11c)$$

where the weekly and monthly climate and biodiversity news indexes are used to account for the intermediate and long-term sentiment as is the case for Equations (10a) - (10e).

Empirical findings

In-sample findings

Table 2 presents the in-sample estimation results for the benchmark models that account for the leverage

effect as well as their augmented variations that incorporate the climate and biodiversity indices. The benchmark LHAR-RV models in Equations (7)-(9) are reported as Models 1, 5, and 9, respectively. Similarly, the LHAR-CLMT models augmented with the climate news index in Equations (10a)-(10c) are reported as Models 2, 6 and 10 and LHAR-BIO models in Equations (10d)-(10f) augmented with the biodiversity news index are reported as Models 3, 7 and 11. The comprehensive models augmented with both the climate and biodiversity news indices in Equations (11a)-(11c) are denoted as LHAR-ALL and reported as Models 4, 8 and 12. The findings for one day, week, and month ahead forecasts are reported in Panels A, B, and C. In line with the evidence in Niu et al. (2024), statistically significant ϕ_1 , ϕ_2 and ϕ_3 estimates justify the use of the leverage component in the forecasting models, implying the presence of leverage effects in green equity volatility patterns at all forecast horizons, with a negative sign for ϕ_1 and ϕ_2 associating negative price fluctuations with higher volatility.

In all panels, we observe a positive and significant jump component within the LHAR-J type model specifications including LHAR-CLMT-J, LHAR-BIO-J, and LHAR-ALL-J specifications. This implies that past realizations of jumps are positively associated with future volatility patterns. This is indeed in line with the finding in Bu et al. (2023) that stock volatility forecasts can be improved by accounting for jumps obtained from high frequency data and by Bégin, Dorion, and Gauthier (2020) that associates idiosyncratic asset risks with jump risk. Signed jumps, however, yield mixed results depending on the forecast horizon with an insignificant effect in the short- (Panel A) and long-term (Panel C); a negative effect in the intermediate horizon (Panel B), specifically for the L-HAR-SJ and LHAR-CLMT-SJ models. Overall, the baseline results support the inclusion of the leverage effect and past jumps as predictors of volatility, thus providing justification for the LHAR specification as the benchmark model.

Examining the predictive influence of the climate and biodiversity news indices reveals a noteworthy finding that climate demonstrates a significant and positive predictive effect on volatility, particularly evident for λ_3 (Climate t-22), across the short, intermediate, and long-term forecast horizons. The positive and statistically significant λ_3 coefficients for the climate news index imply that elevated market sentiment, captured by the media coverage of climate related events, has a significant influence on return volatility in the

Table 2. In-sample estimates of HAR models for various forecasting horizons.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10	Model 11	Model 12
Panel A: 1-day forecast horizon												
α_0	0.0000385	0.0000008	0.0000386	0.000012	0.0000354	-0.000019	0.0000359	0.0000089	0.000038	0.0000004	0.0000384	0.0000118
α_1	0.1487***	0.1459***	0.1471***	0.1443***	0.0985***	0.2005***	0.0979***	0.0942***	0.1445***	0.1417***	0.1423***	0.1395***
α_2	0.3343***	0.3279***	0.3387***	0.3318***	0.3341***	0.3068***	0.3384***	0.3314***	0.336***	0.3295***	0.3408***	0.3339***
α_3	0.3837***	0.3894***	0.3818***	0.3898***	0.3804***	0.311***	0.3786***	0.3867***	0.3849***	0.3907***	0.3832***	0.3912***
ϕ_1	-0.3875***	-0.3885***	-0.3885***	-0.3901***	-0.3952***	-0.3669***	-0.3967***	-0.3988***	-0.4044***	-0.4053***	-0.408***	-0.4096***
ϕ_2	-0.686***	-0.7201***	-0.6724***	-0.7181***	-0.7397***	-0.8809***	-0.7252***	-0.7727***	-0.6857***	-0.7198***	-0.6714***	-0.7172***
ϕ_3	0.9211**	0.8265*	0.9188**	0.8723**	0.9315**	0.7584***	0.9301**	0.8834**	0.9238**	0.8292**	0.9226**	0.876**
β_1					0.1038***	0.0397	0.1021***	0.1041***	0.0483	0.0482	0.0552	0.0553
λ_1		-0.011		-0.011		0.0295		-0.014		-0.012		-0.012
λ_2		-0.025		-0.025		-0.033		-0.022		-0.025		-0.025
λ_3		0.4127***		0.423***		0.4152***		0.4282***		0.4132***		0.4236***
η_1			0.0074	0.0071			0.0042	0.0039			0.0071	0.0068
η_2			-0.056	-0.052			-0.05	-0.045			-0.058	-0.053
η_3			0.0508	0.0083			0.0467	0.0038			0.0521	0.0098
R^2	0.322	0.325	0.322	0.325	0.324	0.3844	0.3253	0.3273	0.3223	0.3253	0.3233	0.3253
Panel A: 1-week forecast horizon												
α_0	0.0000711***	0.0000179	0.0000571***	0.0000209	0.0000689***	0.0000154	0.0000552***	0.0000187	0.0000719***	0.0000186	0.0000574***	0.0000213
α_1	0.0664***	0.0624***	0.0632***	0.0595***	0.0314	0.0265	0.0297	0.025	0.0741***	0.0701***	0.0696***	0.0659***
α_2	0.3605***	0.3508***	0.3714***	0.3614***	0.3603***	0.3506***	0.3712***	0.36113***	0.3575***	0.3479***	0.3687***	0.3588***
α_3	0.4038***	0.4124***	0.3958***	0.4072***	0.4016***	0.4102***	0.3936***	0.4051***	0.4015***	0.41***	0.394***	0.4054***
ϕ_1	-0.301***	-0.3***	-0.3064***	-0.3058***	-0.3064***	-0.3057***	-0.3119***	-0.3117***	-0.2703***	-0.269***	-0.2809***	-0.2803***
ϕ_2	-0.4159***	-0.4643***	-0.3657*	-0.4277***	-0.4533***	-0.5031***	-0.4017***	-0.4654***	-0.4164***	-0.4647***	-0.367*	-0.4289***
ϕ_3	0.8578***	0.7207***	0.7902***	0.7218***	0.8651***	0.7278***	0.7978***	0.7295***	0.8529***	0.7156***	0.7851***	0.7169***
β_1					0.0724***	0.0742***	0.0696***	0.0717***	-0.0883*	-0.0888*	-0.0724	-0.0723
λ_1		-0.002		-0.002		-0.005		-0.004		-0.002		-0.002
λ_2		-0.068		-0.075		-0.066		-0.073		-0.067		-0.074
λ_3		0.6013***		0.5959***		0.6045***		0.5995***		0.6004***		0.5951***
η_1			0.001	0.0003			-0.001	-0.001			0.0014	0.0007
η_2			-0.117***	-0.111***			-0.112***	-0.106***			-0.114***	-0.108***
η_3			0.1645***	0.1092**			0.1617***	0.1062**			0.1628***	0.1073**
R^2	0.521	0.531	0.526	0.535	0.523	0.5335	0.5285	0.5375	0.5225	0.5315	0.5265	0.5355
Panel A: 1-month forecast horizon												
α_0	0.0001291***	0.0000373***	0.0000313*	-0.000023	0.0001281***	0.0000361**	0.0000305*	-0.000024	0.0001289***	0.0000371***	0.0000312*	-0.000023
α_1	0.5555***	0.0483***	0.0535***	0.0475***	0.039**	0.0305*	0.038**	0.0306*	0.053***	0.0461***	0.0505***	0.0446***
α_2	0.1933***	0.1803***	0.1995***	0.1866***	0.1932***	0.1802***	0.1994***	0.18647***	0.1942***	0.1811***	0.2008***	0.1878***
α_3	0.5358***	0.5472***	0.5157***	0.5314***	0.5348***	0.5461***	0.5147***	0.5303***	0.5366***	0.5479***	0.5165***	0.5321***
ϕ_1	-0.1429**	-0.1452**	-0.1419**	-0.1445**	-0.1454**	-0.148**	-0.1444**	-0.1475**	-0.1529**	-0.154**	-0.1538**	-0.1557**
ϕ_2	-0.2737*	-0.3479***	-0.1696	-0.2616*	-0.2914*	-0.3671**	-0.1862	-0.2799*	-0.2735*	-0.3478**	-0.169	-0.261*
ϕ_3	1.0856***	0.8559***	0.6727***	0.5797***	1.0891***	0.8594***	0.6762***	0.5834***	1.0872***	0.8573***	0.675***	0.5818***
β_1					0.0342*	0.0366*	0.032	0.035*	0.0288	0.0251	0.0338	0.0318
λ_1		0.003		0		0.0018		-0.001		0.0028		0
λ_2		-0.014		-0.034		-0.014		-0.034		-0.015		-0.035
λ_3		0.9344***		0.851***		0.936***		0.8527***		0.9347***		0.8513***
η_1			-0.003	-0.004			-0.004	-0.005			-0.003	-0.004
η_2			-0.071***	-0.06***			-0.069***	-0.058**			-0.072***	-0.061***
η_3			0.3839***	0.2899***			0.3826***	0.2884***			0.3847***	0.2908***
R^2	0.542	0.577	0.561	0.587	0.543	0.5786	0.5616	0.5886	0.5425	0.5776	0.5616	0.5876

Note: Table 2 reports the in-sample estimates for the volatility forecasting models. The shaded columns, Models 1, 5, and 9, are the baseline LHAR specifications (Equations 7–9). Models 2, 6, and 10 augment those benchmarks with the NYT Climate News Index (Equations 10a–10c), while Models 3, 7, and 11 add the NYT Biodiversity News Index (Equations 10d–10f). Finally, Models 4, 8, and 12 (labeled LHAR-ALL) include both climate and biodiversity indices (Equations 11a–11c). Panels A, B, and C display coefficients for one-day, one-week, and one-month forecasts, respectively. To conserve space, only coefficient estimates are shown; standard errors are available on request. (*, **, ***) indicate significance at the 10%, 5%, and 1% levels.

market for green equities, suggesting that climate sentiment notably impacts subsequent price fluctuations over various time horizons. Biodiversity sentiment, on the other hand, reveals limited predictive power in the short term implied by insignificant η_1 values. In contrast, we observe that biodiversity related news exerts a delayed impact on market volatility with significant coefficients observed for η_2 and η_3 at the intermediate and long forecast horizons. This finding suggests that biodiversity related news such as changes to the Endangered Species Act in 2019 or the release of reports regarding ecological issues or the extinction of species takes time for the public to process and reflect in financial markets, thus leading to the observed intermediate and long-term effects reported in Panels B and C.

At the same time, examining the intermediate- and long-term forecast horizons (reported in Panels B and C) for η_2 , which has a 1-week lag, shows a dampening effect on volatility. More specifically, when negative biodiversity sentiment arises in the immediate past week, it may encourage market participants to take actions that stabilize or mitigate market fluctuations. This could involve adjustments to investment strategies or increased caution in response to negative biodiversity news. On the other hand, η_3 , which has a 1-month lag, shows a positive impact on volatility, suggesting a more nuanced response over time. Despite initial stabilization efforts in response to negative biodiversity sentiment, market participants may respond positively to initiatives addressing biodiversity concerns. This could include increased investment in sustainable practices or the development of innovative solutions to mitigate biodiversity loss. Ultimately, these actions could lead to a more positive impact on market volatility. Finally, analyzing the predictive impact of climate and biodiversity news indices (Models 4, 8 and 12), it is evident that climate sentiment has a significant and positive predictive effect on volatility over the short, intermediate, and long-term forecast horizons. In contrast, the effect of biodiversity related sentiment is largely concentrated on relatively longer horizons (Panel B and C), consistent with Models 2, 3, 6 and 7.

Accordingly, in-sample results show that both the climate and biodiversity related news have a significant impact on green financial market volatility although the effect of these sentiment proxies can vary depending on the time frames. While climate sentiment has an immediate impact on short-term volatility, biodiversity sentiment has a positive impact on longer-term volatility. This aligns with the efficient market hypothesis, which suggests that market

participants respond quickly to climate sentiment due to climate risks' perceived materiality and urgency. As a result, volatility levels adjust rapidly. On the other hand, the longer-term impact of biodiversity sentiment on volatility suggests a gradual assimilation of biodiversity-related factors into market dynamics. This is because biodiversity risks and opportunities often involve regulatory, social, and technological factors that evolve over extended periods.

Out-of-sample forecasting analysis

Although our in-sample results support the predictive role of climate and biodiversity related news over volatility dynamics in green equities, we are particularly interested in their predictive value out-of-sample to see if augmenting the forecasting models with these predictors can indeed improve out-of-sample forecasts and more importantly, result in economic benefits. Following the literature, to avoid spurious results in the ranking of volatility forecasts, we evaluate the forecasts *via* three loss functions shown to be robust to noise in the volatility proxy. They are the mean squared-error (MSE), heteroskedasticity-adjusted MSE and mean absolute error (MAE) formulated as:

$$MSE = N^{-1} \sum_{t=1}^N (\sigma_t - RV_t)^2, \quad (12a)$$

$$HMSE = N^{-1} \sum_{t=1}^N \left(1 - \frac{RV_t}{\sigma_t}\right)^2, \quad (12b)$$

$$HMAE = N^{-1} \sum_{t=1}^N \left|1 - \frac{RV_t}{\sigma_t}\right|, \quad (12c)$$

where RV_t is the out-of-sample volatility forecast generated by a given model, σ_t is the RV value estimated from the rolling in-sample observations, and N is the length of the out-of-sample evaluation period from November 22, 2018, to June 30, 2022.⁵

The volatility forecasts obtained from competing models are then evaluated using the model confidence set (MCS) of Hansen, Lunde, and Nason (2011). The MCS methodology offers an advantage in our context in that we can use each loss function to rank the competing models and see if their forecast performance significantly differs from each other. The models that remain in the confidence set are then assumed to have similar predictive performance, implying that these models offer superior forecasts at the corresponding confidence level. Considering that the MCS is free from the requirement of a baseline specification that needs to be determined a priori, this procedure

has been a popular choice in out-of-sample forecast evaluation (Niu et al. 2024). Following prior works, MCS statistics are chosen as the range-based (Range) and semi-quadratic (SemiQ) statistics formulated as:

$$T_R = \text{MAX}_{u, v \in M} \frac{|\bar{d}_{i, uv}|}{\sqrt{\text{var}(\bar{d}_{i, uv})}}, \quad T_{SQ} = \text{MAX}_{u, v \in M} \frac{(\bar{d}_{i, uv})^2}{\text{var}(\bar{d}_{i, uv})},$$

$$\bar{d}_{i, uv} = n^{-1} \sum_{t=1}^n d_{i, uv, t},$$
(13)

where $d_{i, uv}$ captures the relative sample loss between the i^{th} and j^{th} models. Note that the p -values are computed *via* bootstrapping program with $p > 0.10$ indicating superior predictive performance.

We report in Table 3 presents the estimated p -values for the Range and SemiQ test statistics, denoted as T_R and T_{SQ} . For ease of reading, we indicate superior models with $p > 0.10$ in bold. The results for one-step, five-step and 22-step forecasts are reported in Panels A, B, and C, respectively. The estimated p -values consistently surpass the 0.1 threshold across both the short and intermediate forecast horizons, reported in Panels A and B. Among the competing models, the highest p -values observed for the LHAR-RV-CLMT, LHAR-RV-BIO, and LHAR-RV-CLMT-BIO variations in Panels A and B. Regarding the models augmented with the jump and sign jump predictors, LHAR-RV-CLMT-J and LHAR-RV-CLMT-SJ emerge with the highest values in Panel A, closely followed by LHAR-RV-BIO-J and LHAR-RV-BIO-SJ. For the intermediate term, while LHAR-RV-J showed the highest p -value among the augmented jump models, the mode variations LHAR-CLMT-J and LHAR-BIO-J, augmented with the climate and biodiversity news indexes respectively, all yield significant p -values exceeding 0.1. Similarly, in the case of the signed jump model variations, LHAR-RV-CLMT-SJ is found to be the best performing model among those with sign jump augmentation, succeeded by the LHAR-RV-BIO-SJ model. Overall, the analysis demonstrates p -values exceeding 0.1 in both the short and intermediate forecast horizons, indicating superior predictive performance of models that incorporate the biodiversity and climate news indexes across all model settings, with particular improvements observed in models featuring jump and sign jump augmentations.

Economic implications

In the final step of our analysis, we examine whether the predictive information captured by climate and

Table 3. Out-of-sample forecast accuracy based on the model confidence set (MCS).

	Range			SemiQ		
	MSE	HMSE	HMAE	MSE	HMSE	HMAE
Panel A: 1-day forecast horizon						
LHAR-RV	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
LHAR-RV-CLMT	0.9361	0.2504	0.014	0.9361	0.191	0.0101
LHAR-RV-BIO	0.5891	0.5123	0.0733	0.6989	0.5123	0.0435
LHAR-RV-CLMT-BIO	0.5891	0.4224	0.0733	0.6989	0.3404	0.0435
LHAR-RV-J	0.2566	1.0000	1.0000	0.2467	1.0000	1.0000
LHAR-RV-CLMT-J	1.0000	0.9001	0.1815	1.0000	0.9042	0.1221
LHAR-RV-BIO-J	0.2566	0.9001	0.1815	0.2467	0.9042	0.1221
LHAR-RV-CLMT-BIO-J	0.2566	0.6017	0.1815	0.2467	0.5676	0.1059
LHAR-RV-SJ	0.0027	1.0000	1.0000	0.0081	1.0000	1.0000
LHAR-RV-CLMT-SJ	1.0000	0.2496	0.0135	1.0000	0.1965	0.0103
LHAR-RV-BIO-SJ	0.8334	0.5377	0.0742	0.8334	0.5377	0.0422
LHAR-RV-CLMT-BIO-SJ	0.7752	0.4233	0.0135	0.8286	0.3478	0.0103
Panel B: 1-week forecast horizon						
LHAR-RV	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
LHAR-RV-CLMT	0.9172	0.1842	0.0087	0.9172	0.1462	0.0064
LHAR-RV-BIO	0.4461	0.1842	0.0087	0.5339	0.1462	0.0064
LHAR-RV-CLMT-BIO	0.4405	0.1842	0.0087	0.5075	0.1462	0.0064
LHAR-RV-J	1.0000	1.0000	1.0000	0.9799	0.1511	0.0066
LHAR-RV-CLMT-J	0.9799	0.1947	0.0096	0.5472	0.1511	0.0066
LHAR-RV-BIO-J	0.3571	0.1947	0.0096	0.5472	0.1511	0.0066
LHAR-RV-CLMT-BIO-J	0.3571	0.1947	0.0096	0.6989	0.3404	0.0435
LHAR-RV-SJ	0.0023	1.0000	1.0000	0.0053	1.0000	1.0000
LHAR-RV-CLMT-SJ	1.0000	0.1746	0.0084	1.0000	0.1408	0.0064
LHAR-RV-BIO-SJ	0.6644	0.1746	0.0084	0.6644	0.1408	0.0064
LHAR-RV-CLMT-BIO-SJ	0.4531	0.1746	0.0084	0.5858	0.1408	0.0064
Panel C: 1-month forecast horizon						
LHAR-RV	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
LHAR-RV-CLMT	0.6645	0.0735	0.0055	0.6645	0.0735	0.0055
LHAR-RV-BIO	0.0000	0.0516	0.0018	0.0000	0.0373	0.0005
LHAR-RV-CLMT-BIO	0.0000	0.0516	0.0005	0.0000	0.0373	0.0003
LHAR-RV-J	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
LHAR-RV-CLMT-J	0.8257	0.0732	0.0056	0.8257	0.0732	0.0056
LHAR-RV-BIO-J	0.0251	0.0517	0.0020	0.0616	0.0372	0.0005
LHAR-RV-CLMT-BIO-J	0.0251	0.0517	0.0005	0.0616	0.0372	0.0003
LHAR-RV-SJ	0.0011	1.0000	1.0000	0.0009	1.0000	1.0000
LHAR-RV-CLMT-SJ	1.0000	0.0736	0.0054	1.0000	0.0736	0.0054
LHAR-RV-BIO-SJ	0.0452	0.0516	0.0018	0.0452	0.0373	0.0005
LHAR-RV-CLMT-BIO-SJ	0.0127	0.0516	0.0005	0.0196	0.0373	0.0003

Note: Table 3 shows p -values from the Model Confidence Set procedure using the Range and SemiQ statistics (TR and TSQ). Entries in bold have p -values above 0.10. Panels A, B, and C present results for 1-step, 5-step, and 22-step realized-volatility forecasts, respectively. In each panel, the shaded rows display the baseline LHAR model, which is then tested against its extensions-LHAR-CLMT, LHAR-BIO, and LHAR-ALL, that incorporate the NYT climate and biodiversity news indices. MSE, HMSE, and HMAE refer to mean squared error, heteroskedasticity-adjusted MSE, and mean absolute error, respectively.

biodiversity sentiment can be utilized in an investment setting. Considering that volatility is a key parameter in the computation of optimal investment positions, accurate prediction of volatility becomes a key concern in asset allocation decisions. To that end, following the literature (e.g. Neely et al. 2014), we consider an investment strategy and examine if the volatility forecasts from the models augmented by the climate series can improve the risk and return profile for a passive investor. In this setting, we assume that the investor allocates the portfolio between the green asset (green ETF in our case) and risk-free Treasury securities based on the following utility function:

$$\begin{aligned}
U(R_p) &= E(R_p) - \frac{\gamma}{2} \text{Var}(R_p) \\
&= w\theta(r_t + r_t^f) + (1 - w_t)r_t^f - \frac{\gamma}{2} w_t^2 \theta_t^2 \sigma_t^2,
\end{aligned} \tag{14}$$

where γ captures risk aversion, w_t is the portfolio weight and r_t is the excess return on the green ETF, respectively. The risk-free return is denoted by r_t^f while θ represents the leverage ratio. Following this setting, we can formulate the portfolio return as $R_p = w_t\theta_t(r_t + r_t^f) + (1 - w_t)r_t^f$, and its variance as $\text{Var}(R_p) = w_t^2\theta_t^2\sigma_t^2$. Next, we obtain the utility maximizing portfolio position on the green ETF as:

$$w_t^* = \frac{\theta r_{t+1} + (\theta - 1)r_{t+1}^f}{\gamma\theta^2\sigma_{t+1}^2}, \tag{15}$$

where r_{t+1} is the mean of in-sample data proxied by the historical average return assuming an efficient market setting (Campbell and Thompson 2008) and σ_{t+1}^2 is the out-of-sample realized volatility forecast for the green ETF on day $t + 1$. In this setting, volatility forecasts that we generate based on alternative models described earlier become the key parameter that determine the optimal position in the green asset. Note that the leverage ratio is set to unity following the literature (e.g. Rapach, Strauss, and Zhou 2010), implying the restriction on margin trades, thus resulting in the return for the portfolio on day $t + 1$ formulated as $R_{t+1} = w_t^*r_{t+1} + r_{t+1}^f$. To evaluate the economic implications of the volatility forecasts, the first measure we use is the certainty equivalent return (CER) that we formulate as $CER = \bar{R}_p - 0.5\gamma\sigma_p^2$, where $\bar{R}_p$ and σ_p^2 are the portfolio's out-of-sample mean and variance, respectively. In line with Rapach, Strauss, and Zhou (2010) and others in the literature, the risk aversion value is set to three. Additionally, we evaluate the portfolios using the Sharpe ratio, which is the ratio of the portfolio's out-of-sample excess return to its volatility.

To enhance clarity regarding the portfolio formation process based on volatility forecasts, we provide a step-by-step description of our dynamic investment strategy. At each out-of-sample period, the investor allocates wealth between the green equity ETF and a risk-free asset. The optimal weight assigned to the green ETF is determined by maximizing expected utility under mean-variance preferences, where the model's forecasted volatility and the historical average return are used as inputs for expected risk and return, respectively. The formula for the optimal weight is presented in Equation (15), with all relevant variables, historical average return, risk-free rate, risk aversion

Table 4. Economic analysis of out of sample forecasts.

	$h = 1\text{-day}$		$h = 1\text{-week}$		$h = 1\text{-month}$	
	CER	Sharpe	CER	Sharpe	CER	Sharpe
LHAR-RV	0.0221	2.6545	0.0233	3.2333	0.0225	3.6916
LHAR-RV-CLMT	0.0227	2.7334	0.0231	3.3280	0.0216	3.8027
LHAR-RV-BIO	0.0214	2.4138	0.0192	2.3704	0.0190	3.9354
LHAR-RV-CLMT-BIO	0.0206	2.3236	0.0106	1.8702	0.0180	5.0551
LHAR-RV-J	0.0202	2.4991	0.0225	3.1310	0.0222	3.6480
LHAR-RV-CLMT-J	0.0181	2.1590	0.0224	3.2294	0.0214	3.7708
LHAR-RV-BIO-J	0.0195	2.2992	0.0184	2.3299	0.0190	3.9118
LHAR-RV-CLMT-BIO-J	0.0194	2.2713	0.0096	1.8297	0.0179	5.0046
LHAR-RV-SJ	0.0225	2.6732	0.0227	3.1717	0.0226	3.7169
LHAR-RV-CLMT-SJ	0.0229	2.7353	0.0226	3.2879	0.0217	3.8027
LHAR-RV-BIO-SJ	0.0209	2.3532	0.0204	2.5155	0.0191	3.9130
LHAR-RV-CLMT-BIO-SJ	0.0203	2.2983	0.0115	1.9225	0.0173	4.5863

Note: CER and Sharpe denote the out-of-sample certainty equivalent return and Sharpe ratio, respectively. Shaded rows display the baseline LHAR model; boldface entries highlight the LHAR-type variant that delivers the highest economic value relative to that benchmark.

parameter, and the model's volatility forecast clearly defined. The realized portfolio return is then calculated as the weighted sum of the green ETF and risk-free asset returns. Over the evaluation period, we assess the economic value of each model's forecasts by computing both the Sharpe ratio and the certainty equivalent return (CER).

The portfolio measures obtained using the volatility forecasts from each competing model are reported in Table 4. In the table, we denote the benchmark model in shaded rows and bold numbers are used for cases where the alternative model that yields the greatest economic benefit in comparison to the benchmark. We observe at the 1-day forecast horizon that the LHAR-RV-CLMT and LHAR-RV-CLMT-SJ models yield the highest CER and Sharpe ratios, consistent with the earlier evidence that the predictive power of climate news sentiment over volatility is largely concentrated on short forecast horizons as market participants respond quickly to climate sentiment due to climate risks' perceived materiality and urgency. We find that incorporating climate sentiment into trading decisions enables investors to adeptly navigate short-term market movements and exploit inefficiencies for profit. Moreover, leveraging climate sentiment in trading strategies not only enhances potential returns but also facilitates more effective risk management by keeping traders informed about environmental trends that may influence asset prices in the short term.

For the 1-week forecast horizon, the highest Sharpe ratio values are observed across all climate-based forecasting models, specifically the LHAR-RV-CLMT, LHAR-RV-CLMT-J, and LHAR-RV-CLMT-SJ model variations, suggesting that integrating climate sentiment alongside traditional forecasting methods can augment performance metrics, providing investors with valuable insights into short- and intermediate-

term market dynamics. Looking ahead to the 1-month forecast horizon, notably high Sharpe ratio values observed in models like LHAR-RV-CLMT-BIO underscore the advantages of integrating environmental sentiment factors into longer-term investment strategies, aligning with sustainability objectives. While climate sentiment models may excel in capturing shorter-term performance, incorporating both climate and biodiversity sentiment factors proves beneficial for longer-term investment horizons. Overall, the analysis of economic implications of the predictive relationships observed in the forecasting analysis highlights the importance of adopting a diversified approach to model selection and risk management, considering the broader context and potential tradeoffs involved. The trend of integrating environmental sentiment factors into investment strategies for green equities reflects increasing awareness of sustainability and climate-related risks. By diversifying across multiple models and incorporating environmental sentiment, investors can mitigate risks associated with model uncertainty and maximize opportunities for consistent performance across varying market conditions.

Extension to directional volatility

Good and bad realized volatility

Although standard risk models assume away potential asymmetries in investors' risk preferences when it comes to making investment decisions, in practice, differentiating between upside and downside risk becomes critical to address investors' preference toward loss aversion and more importantly, to value securities with option like features. This is especially important for green equities as these firms deal with new technologies that are characterized by real option like payoffs. Against this backdrop, we extend our analysis to measures of asymmetric risk and further explore the predictive value of climate and biodiversity sentiment over measures of good and bad volatility in green equity returns. To that end, following Gong and Lin (2017), we formulate good (GV) and bad (BV) realized volatility as $GV_t = \sum_{j=1}^M I\{r_{t,j} > 0\} r_{t,j}^2$ and $BV_t = \sum_{j=1}^M I\{r_{t,j} < 0\} r_{t,j}^2$, respectively, with the indicator function I is used to differentiate between signed returns. Next, the forecasting models with and without the climate/biodiversity sentiment proxies are evaluated for each directional volatility measure.

In the case of good volatility (GV) forecasts, four competing models are formulated, i.e. HAR-GV, LHAR-GV, LHAR-GV-CLMT, and LHAR-GV-BIO and LHAR-GV-CLMT-BIO, as

$$\overline{GV}_{t+H} = \alpha_0 + \alpha_1 GV_t + \alpha_2 GV_{t-5,t} + \alpha_3 GV_{t-22,t} + \varepsilon_{t+H}, \quad (16a)$$

$$\overline{GV}_{t+H} = \alpha_0 + \alpha_1 GV_t + \alpha_2 GV_{t-5,t} + \alpha_3 GV_{t-22,t} + \phi' R_t + \varepsilon_{t+H}, \quad (16b)$$

$$\overline{GV}_{t+H} = \alpha_0 + \alpha_1 GV_t + \alpha_2 GV_{t-5,t} + \alpha_3 GV_{t-22,t} + \phi' R_t + \lambda' Clmt_t + \varepsilon_{t+H}, \quad (16c)$$

$$\overline{GV}_{t+H} = \alpha_0 + \alpha_1 GV_t + \alpha_2 GV_{t-5,t} + \alpha_3 GV_{t-22,t} + \phi' R_t + \lambda' Bio_t + \varepsilon_{t+H}, \quad (16d)$$

$$\overline{GV}_{t+H} = \alpha_0 + \alpha_1 GV_t + \alpha_2 GV_{t-5,t} + \alpha_3 GV_{t-22,t} + \phi' R_t + \lambda' Clmt_t + \eta' Bio_t + \varepsilon_{t+H}, \quad (16e)$$

where GV_b is the positive realized semi-variance on day t ; and $GV_{t,t-5} = (GV_t + GV_{t-1} + \dots + GV_{t-4})/5$ and $GV_{t,t-22} = (GV_t + GV_{t-1} + \dots + GV_{t-22})/22$ are the weekly and monthly series, respectively. The predictive role of climate sentiment for good volatility forecasts is examined *via* models LHAR-GV-CLMT, LHAR-GV-BIO and LHAR-GV-CLMT-BIO shown in Equations (16c) and (16e) and the same procedure is applied for bad volatility *via* an additional set of models including HAR-BV, LHAR-BV, LHAR-BV-CLMT, LHAR-BV-BIO, and LHAR-BV-CLMT-BIO.

In sample analysis

The estimated coefficients for 1-day, 1-week, and 1-month ahead forecasts are reported in Table 5 Panels A, B, and C, respectively. Models (1)-(5) represent the good volatility forecasts for HAR-GU, LHAR-GU, LHAR-GU-CLMT, LHAR-GU-BIO and LHAR-GU-ALL, shown in Equations (16a)-(16e), respectively. Similarly, Models (6)-(10) represent those for bad volatility that we formulate in a similar fashion. Consistent with the earlier results, incorporating the leverage term helps to improve forecast accuracy of the models with the LHAR models generally outperforming their HAR-based alternatives, as evidenced by their higher R-square values. Incorporating the leverage term in the model is further justified by the estimated coefficients for daily leverage effects that are significant at the 1% level. Furthermore, we note even higher R-square values upon expanding models incorporating climate and

Table 5. In-sample estimates for good and bad volatility.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10
Panel A: 1-day forecast horizon										
α_0	0.00006***	0.00003	−0.00001	0.00003	0.00001	0.00004***	0.00005**	0.00001	0.00004	0.00002
α_1	0.2949***	0.2746***	0.2695***	0.2715***	0.2665***	0.311***	0.2793***	0.2732***	0.2766***	0.2708***
α_2	0.5597***	0.5483***	0.5334***	0.5602***	0.5442***	0.7906***	0.7771***	0.7666***	0.7812***	0.7704***
α_3	0.8276***	0.8581***	0.8716***	0.8529***	0.8717***	0.6549***	0.7095***	0.7178***	0.7074***	0.7197***
ϕ_1		−0.43***	−0.429***	−0.431***	−0.431***		−0.342**	−0.345***	−0.342**	−0.345***
ϕ_2		−0.847***	−0.882***	−0.833***	−0.881***		−0.545*	−0.578*	−0.533*	−0.577*
ϕ_2		0.8525**	0.7523*	0.8656**	0.8147*		0.9913**	0.8961**	0.9694**	0.923**
λ_1			−0.012		−0.011			−0.011		−0.011
λ_2			−0.031		−0.03			−0.022		−0.022
λ_3			0.4352***		0.4494***			0.4101***		0.4155***
η_1				0.0075	0.0072				0.0086	0.0082
η_2				−0.066	−0.061				−0.044	−0.039
η_3				0.0484	0.0033				0.0527	0.0112
R^2	0.308	0.318	0.320	0.318	0.321	0.315	0.320	0.322	0.320	0.322
Panel B: 1-week forecast horizon										
α_0	0.00008***	0.00009***	0.00003	0.00007**	0.00003	0.00007***	0.00011***	0.00005*	0.00008**	0.00004
α_1	0.1172**	0.1098**	0.1015**	0.0993**	0.0919*	0.1937***	0.1893***	0.1798***	0.1766***	0.1683***
α_2	0.5652***	0.5608***	0.5323***	0.6***	0.5711***	0.7212***	0.7268***	0.7038***	0.7494***	0.7264***
α_3	0.8623***	0.883***	0.9089***	0.8528***	0.8851***	0.6966***	0.7236***	0.7426***	0.7086***	0.733***
ϕ_1		−0.125	−0.115	−0.128	−0.117		−0.066	−0.059	−0.068	−0.062
ϕ_2		−0.368	−0.426	−0.292	−0.364		−0.026	−0.084	0.0456	−0.023
ϕ_2		0.6876	0.5052	0.5745	0.4752		0.7899*	0.6189	0.647	0.556
λ_1			0.0365		0.0347			0.0359		0.0334
λ_2			−0.188**		−0.2**			−0.180*		−0.194**
λ_3			0.8019***		0.778***			0.7615***		0.7322***
η_1				0.0282	0.0268				0.0273	0.0258
η_2				−0.228***	−0.221***				−0.206***	−0.2***
η_3				0.2842***	0.222***				0.2876***	0.2315***
R^2	0.242	0.244	0.252	0.251	0.259	0.251	0.253	0.260	0.260	0.266
Panel C: 1-month forecast horizon										
α_0	0.00011***	0.00016***	0.00003	−0.00006	−0.0013***	0.0001***	0.00018***	0.00004	−0.00005	−0.00012***
α_1	0.154***	0.1535***	0.1318**	0.149***	0.1324**	0.1807***	0.1782***	0.1575**	0.1703***	0.1552**
α_2	0.0185	0.0309	0.0052	0.0483	0.0242	−0.086	−0.071	−0.089	−0.062	−0.078
α_3	1.3346***	1.3461***	1.3703***	1.257***	1.2936***	1.5037***	1.5452***	1.5538***	1.4705***	1.4914***
ϕ_1		−0.135	−0.135	−0.126	−0.128		−0.066	−0.074	−0.058	−0.065
ϕ_2		0.1318	0.0467	0.3432	0.2353		0.0747	−0.004	0.2801	0.1802
ϕ_2		0.8315	0.4987	−0.085	−0.183		1.3542***	1.0183**	0.4058	0.3036
λ_1			0.0845*		0.0743*			0.0855*		0.0746*
λ_2			0.1247		0.0806			0.1181		0.0726
λ_3			1.1699***		0.9634***			1.1537***		0.9427***
η_1				0.0009	−0.001				0.0023	0.000
η_2				−0.107*	−0.087				−0.099*	−0.08
η_3				0.7981***	0.6491***				0.8063***	0.6633***
R^2	0.179	0.181	0.210	0.210	0.229	0.183	0.188	0.217	0.219	0.236

Note: Panels A, B, and C display coefficient estimates for 1-day, 1-week, and 1-month ahead forecasts, respectively. Within each panel, Models (1)–(5) correspond to the HAR-GV, LHAR-GV, LHAR-GV-CLMT, LHAR-GV-BIO, and LHAR-GV-ALL specifications for good volatility (Equations 16a–16e), while Models (6)–(10) display specifications for bad volatility. Only point estimates are shown; standard errors are available upon request. (*, **, ***) indicate significance at the 10%, 5%, and 1% levels.

biodiversity sentiment. Notably, the LHAR-GV-CLMT-BIO and LHAR-BV-CLMT-BIO models consistently demonstrate the highest R-square values across all forecast horizons, supporting the inclusion of the climate/biodiversity related predictors in the forecasting models.

We observe that climate sentiment significantly predicts volatility in green equities at all forecast horizons, consistently for both directional volatility models, as indicated by the significant λ_3 values in Panels A, B, and C. Biodiversity sentiment is also found to predict green equity volatility in the intermediate and long term for both directional models, as demonstrated by significant η_3 values in Panels B and C. The positive estimated coefficients for λ_3 and η_3 suggest

that heightened sentiment regarding climate and biodiversity related concerns correlates with increased return volatility in either market direction in the short term, associating sentiment with market fluctuations in both the positive and negative directions. When considering both climate and biodiversity sentiment, we observe short-, intermediate-, and long-term spillover effects on green equity volatility, as depicted in Panels A, B, and C. The predictive power of climate related sentiment over green ETF volatility underscores the importance of considering environmental factors in financial decision-making. Traders and investors can leverage sentiment analysis to anticipate better and manage volatility in green ETF markets,

thereby optimizing their investment strategies and risk management approaches. Moreover, the observed predictive roles of climate and biodiversity sentiment highlight the interconnectedness between environmental trends and market dynamics. This suggests that shifts in sentiment toward climate and biodiversity issues can have lasting impacts on asset prices, influencing investment decisions across various time horizons. As such, market participants need to stay attuned to evolving environmental sentiment trends to navigate green ETF markets and capitalize on emerging opportunities effectively.

From a regulatory perspective, the analysis of asymmetric volatility forecasts emphasizes the importance of incorporating environmental considerations into financial market oversight and policymaking. Regulators may need to enhance monitoring and surveillance efforts to detect and mitigate potential risks associated with fluctuations in sentiment-driven volatility in green equities. Additionally, policymakers may explore initiatives to promote transparency and disclosure of environmental data, fostering greater market efficiency and investor confidence. Overall, the findings underscore the growing significance of environmental factors in shaping financial market dynamics and highlight the need for a holistic approach to sustainable investing. By integrating environmental sentiment analysis into investment strategies and regulatory frameworks, market participants and policymakers can work toward building more resilient and sustainable financial markets that align with broader environmental goals.

Economic analysis of asymmetric predictability

We report in Table 6 the results from out-of-sample performance of the benchmark models against the competing models that incorporate the climate proxies. The benchmark models for good and bad volatility (HAR-GV, HAR-BV) are reported in shaded rows, and we compare their forecasting performance against the LHAR-CLMT, LHAR-BIO and LHAR-ALL models using the model confidence set (MCS) procedure described earlier. We present in the table the p -values for the Range (T_R) and SemiQ (T_{SQ}) test statistics computed for the mean squared-error (MSE), heteroskedasticity-adjusted MSE (HMSE) and mean absolute error (HMAE). Note that p -values that exceed 0.1 (shown in bold) indicate superior performance of the corresponding model.

In Panel A, the p -values are found to consistently exceed 0.1 for models incorporating climate and

Table 6. Out-of-sample forecast accuracy for good and bad volatility.

	Range			SemiQ		
	MSE	HMSE	HMAE	MSE	HMSE	HMAE
Panel A: 1-day forecast horizon						
HAR-GV	0.4435	0.1599	0.0059	0.6188	0.1514	0.0165
LHAR-GV	0.9927	0.8990	1.0000	0.9927	0.8990	1.0000
LHAR-GV-CLMT	1.0000	0.1599	0.0059	1.0000	0.1370	0.0091
LHAR-GV-BIO	0.6154	1.0000	0.1216	0.8366	1.0000	0.1216
LHAR-GV-ALL	0.9869	0.4751	0.1083	0.9923	0.4152	0.0681
HAR-BV	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
LHAR-BV	0.7099	0.8727	0.9356	0.6597	0.8727	0.9356
LHAR-BV-CLMT	0.7099	0.1609	0.0130	0.6597	0.1042	0.0056
LHAR-BV-BIO	0.5884	0.1609	0.0130	0.6094	0.1042	0.0067
LHAR-BV-ALL	0.5884	0.1609	0.0130	0.6094	0.1042	0.0067
Panel B: 1-week forecast horizon						
HAR-GV	0.3329	1.0000	0.4870	0.5211	1.0000	0.4870
LHAR-GV	1.0000	0.5629	1.0000	1.0000	0.5629	1.0000
LHAR-GV-CLMT	0.8907	0.1700	0.0129	0.8907	0.1574	0.0106
LHAR-GV-BIO	0.0435	0.1066	0.0065	0.1088	0.0853	0.0021
LHAR-GV-ALL	0.0435	0.1066	0.0065	0.1188	0.0853	0.0021
HAR-BV	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
LHAR-BV	0.0995	0.0972	0.0178	0.1417	0.0594	0.0178
LHAR-BV-CLMT	0.0995	0.0972	0.0091	0.1417	0.0594	0.0045
LHAR-BV-BIO	0.0057	0.0749	0.0035	0.0069	0.0553	0.0009
LHAR-BV-ALL	0.0187	0.0749	0.0035	0.0143	0.0553	0.0009
Panel C: 1-month forecast horizon						
HAR-GV	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
LHAR-GV	0.2898	0.0854	0.0194	0.2304	0.0573	0.0194
LHAR-GV-CLMT	0.2898	0.0854	0.0035	0.2304	0.0573	0.0018
LHAR-GV-BIO	0.0076	0.0620	0.0011	0.0036	0.0238	0.0002
LHAR-GV-ALL	0.0024	0.0620	0.0005	0.0012	0.0238	0.0002
HAR-BV	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
LHAR-BV	0.1424	0.0980	0.0128	0.1108	0.0602	0.0128
LHAR-BV-CLMT	0.1424	0.0980	0.0033	0.1108	0.0602	0.0013
LHAR-BV-BIO	0.0079	0.0602	0.0011	0.0029	0.0242	0.0002
LHAR-BV-ALL	0.0057	0.0602	0.0005	0.0014	0.0275	0.0002

Note: Table 6 reports Model Confidence Set (MCS) p -values using the Range (TR) and SemiQ (TSQ) statistics. Values exceeding 0.10 are highlighted in bold. Panels A, B, and C correspond to 1-step, 5-step, and 22-step realized-volatility forecast horizons, respectively. In each panel, the shaded rows show the baseline HAR-GV (good volatility) and HAR-BV (bad volatility) models, which are then compared to the expanded LHAR-CLMT, LHAR-BIO, and LHAR-ALL specifications that include the NYT climate and biodiversity news indices. MSE, HMSE, and HMAE denote the mean squared error, heteroskedasticity-adjusted MSE, and mean absolute error.

biodiversity sentiment indexes. This suggests significant short-term information captured by both sentiment proxies, consistently for both directional volatility series in green equity returns. Specifically, we find, for positive volatility, that the LHAR-GV-CLMT model performs best, highlighting the predictive value of sentiment in the short run, consistent with our earlier findings for symmetric volatility forecasts. In contrast, for negative volatility, the HAR-BV model is found to be the most effective, followed by the LHAR-BV-CLMT model. For the longer forecast horizons reported in Panels B and C, we again observe that the models that incorporate the climate sentiment yield the best performance among the competing models with the LHAR-GV-CLMT and LHAR-BV-CLMT models exhibiting the highest performance along with the benchmark HAR-BV and HAR-GV models.

Drawing parallels with Baur's argument for gold (2012), our findings indicate that positive developments in the green energy sector are likely to amplify volatility in green equities more than negative volatility. One reason for this result could be due to investors' overreaction to news or overconfidence in the future prospects of these assets. Positive shocks, like regulatory support or technological breakthroughs, often increase investor interest in green equities, leading to higher market reactions and volatility. Conversely, negative developments, such as setbacks in renewable energy projects, may lower investor confidence, resulting in reduced volatility. Thus, the predictability of volatility dynamics in green equities due to sentiment may vary depending on market dynamics and underlying sentiment in the green energy sector.

To examine the investment implications of the predictive analysis presented above, we report in Table 7 the cumulative returns and CERs for the portfolio that we obtain using the out-of-sample volatility forecasts. Our analysis shows that models focusing on good volatility tend to yield higher Sharpe and CER estimates for the 1-day forecast horizon compared to models focusing on bad volatility. This indicates the presence of potential inefficiencies in how investors process good news in the emerging green technologies so that accurate forecasts of market fluctuations in response to good news can help extract economic gains more so than the case for bad news. At the short forecast horizons, however, while the HAR-GV model emerges as the top performer, followed by LHAR-GV, LHAR-GV-CLMT, LHAR-GV-ALL models, the HAR-BV model is found to exhibit the highest economic performance for bad volatility, followed by LHAR-BV-CLMT and LHAR-BV-ALL models. In that regard, the predictive power of climate related

sentiment proxies does not necessarily translate to economic gains in short run, which could possibly be due to the market's inability to process the information efficiently in the short run for profitable investment strategies.

In the case of the intermediate and long forecast horizons, however, we find that the out-of-sample economic benefits from incorporating the climate sentiment indexes in volatility models are largely concentrated on bad volatility. Although the benchmark models yield the largest economic gains for good volatility, followed by the LHAR-type models, for bad volatility, the LHAR-BV-CLMT model stands out with the highest reward to risk ratios (Sharpe) at the 1-week horizon, while the LHAR-BV-ALL model yields the highest risk-adjusted returns, followed by HAR-BV, LHAR-BV-CLMT, LHAR-BV, and LHAR-BV-BIO models. Overall, our analysis reveals that the predictive power of climate/biodiversity related sentiment indexes can be particularly beneficial for bad volatility forecasts, which is an important consideration for the accurate measurement of downside risk measures in these assets. Thus, we conclude that by incorporating climate and biodiversity sentiment into investment strategies, investors can optimize risk-adjusted returns and potentially capitalize on market opportunities while promoting sustainability objectives.

Overall, our findings show that climate-based models effectively capture short-term market performance due to the immediate and visible effects of climate change. Rapid shifts in environmental conditions, such as temperature fluctuations, extreme weather events, and climate-related policy changes, can have direct and noticeable consequences for industries and financial markets (Stern 2007). These swift and tangible impacts make climate sentiment a key factor in influencing short-term performance, as investors react quickly to both positive and negative news related to climate change. This also shows that investors respond efficiently to observable climate events as they unfold, and this brings about predictability in subsequent volatility patterns.

In contrast, biodiversity-based models offer an advantage over longer forecast horizons. According to Flammer, Giroux, and Heal (2025), biodiversity finance projects typically follow a J-curve pattern, where initial investments yield limited returns, but substantial financial and ecological benefits materialize over time. This delayed return is due to the gradual nature of biodiversity impacts, which take time to manifest and deliver measurable results. Over the long term, both biodiversity and financial outcomes tend to

Table 7. Economic values for the asymmetric HAR models at various forecasting horizons.

	<i>h</i> = 1-day		<i>h</i> = 1-week		<i>h</i> = 1-month	
	CER	Sharpe	CER	Sharpe	CER	Sharpe
HAR-GV	0.0325	3.5462	0.0297	3.8534	0.0265	4.2216
LHAR-GV	0.0242	2.7639	0.0259	3.7754	0.0230	4.0495
LHAR-GV-CLMT	0.0232	2.6786	0.0252	3.6878	0.0207	3.1450
LHAR-GV-BIO	0.0060	1.6547	0.0045	1.6082	0.0042	1.6766
LHAR-GV-ALL	0.0210	2.2888	0.0228	3.2320	0.0140	3.3159
HAR-BV	0.0281	3.1112	0.0269	3.5184	0.0253	3.9807
LHAR-BV	0.0177	2.3397	0.0233	3.4210	0.0206	3.5289
LHAR-BV-CLMT	0.0203	2.5820	0.0240	3.7111	0.0212	3.6360
LHAR-BV-BIO	0.0123	2.0726	0.0113	1.8644	0.0156	2.9237
LHAR-BV-ALL	0.0190	2.3751	0.0221	3.4787	0.0148	4.7538

Note: Sharpe and CER report the portfolio's cumulative return and certainty equivalent return over the out-of-sample horizon. The shaded rows display the baseline HAR-GV and HAR-BV models (for good and bad volatility, respectively). Entries in bold highlight the HAR-type variant that achieves the highest economic value relative to its benchmark.

align, as the ecological benefits of the project enhance financial returns. As a result, the long-term volatility in biodiversity finance is less driven by sudden shifts and more by gradual ecological changes, which may not always be immediately apparent. This long-term volatility could manifest in the form of slower, more persistent shifts in performance as ecosystems recover or degrade over time, reflecting the complex, evolving relationship between financial returns and biodiversity outcomes. Therefore, while short-term volatility is closely tied to climate events, long-term volatility is more closely associated with the gradual, sometimes unpredictable, changes in biodiversity, which can compound over time and influence overall performance in subtle but significant ways. Nevertheless, combining both factors allows investors to make more informed decisions that reflect the full spectrum of environmental risks, improving their ability to forecast long-term performance and optimize their portfolios accordingly.

Robustness checks

As mentioned earlier, our main analysis utilized the Invesco Global Clean Energy ETF to capture equity market dynamics in green equities as this ETF is popularly used in the literature as a proxy for green stocks (e.g. Bouri, Iqbal, and Klein 2022). To check if our inferences also hold for alternative proxies of green equities, we also analyzed intraday data for two alternative green ETFs. The first is the Invesco Wilder Hill Clean Energy ETF (PBW) that aims to replicate the WilderHill Clean Energy Index. The index includes U.S. listed companies that are engaged in the advancement of cleaner energy and conservation, particularly focusing on greener and generally renewable sources of energy and technologies that facilitate cleaner energy. The second alternative ETF we examined for robustness is the iShares Global Clean Energy ETF (ICLN) that is designed to track the performance of approximately 100 global clean energy-related companies included in the S&P Global Clean Energy Index™. ICLN includes companies engaged in various sub-sectors such as wind power, solar power, and other renewable sources, providing targeted access to clean energy stocks from around the world. Considering these two ETFs in our robustness checks, thus, allows to see if our findings hold for green equity proxies at the U.S. and global level.

Our analysis for these alternative ETFs generally confirms our findings in that the out-of-sample predictive power of climate/biodiversity related sentiment

proxies can be leveraged to generate economic benefits. We observe that models incorporating the biodiversity news index particularly, such as LHAR-RV-CLMT-BIO, LHAR-RV-CLMT-BIO-J, and LHAR-RV-ALL-SJ, yield higher Sharpe values at both the short- and intermediate-term forecast horizons, while like the LHAR-RV-ALL model yields the highest risk adjusted returns at the 1-month forecast horizon. This finding underscores the advantages of integrating environmental sentiment factors into longer-term investment strategies, aligning with sustainability objectives. These results affirm that while climate sentiment models may better capture shorter-term performance, incorporating both climate and biodiversity sentiment factors proves beneficial for longer-term investment horizons. Due to space considerations, we do not include the results for these ETFs in the paper and only report them in an online Appendix that is available upon request.

Finally, we would like to note that while our analysis establishes robust predictive relationships between climate and biodiversity sentiment and future volatility, we acknowledge that strict causal inference remains challenging to assess in financial market research. The observed predictive power of sentiment indices may, in part, reflect the influence of unobserved factors such as anticipated regulatory changes or technological developments that simultaneously affect both media sentiment and market volatility. Although our empirical strategy, which combines in-sample and out-of-sample validation, provides strong evidence of a persistent link, we cannot fully rule out potential endogeneity or omitted variable concerns. However, the dual approach of combining in-sample and out-of-sample evidence adopted in our application, which is consistent with the volatility forecasting literature (e.g., Andersen and Bollerslev 1998; Corsi 2008;; Niu et al. 2024), demonstrates that our findings are robust. While we acknowledge that establishing strict causality in financial markets is inherently challenging due to the complex interplay of news, sentiment, and market behavior, our comprehensive empirical strategy provides a high degree of confidence in the reliability and economic significance of our results.

Conclusion

Achieving the U.N. sustainability goals requires not only the coordination among policy makers to realize climate mitigation mandates, but also financial markets to facilitate the financing and development of supporting industries and technologies toward a net-

zero carbon footprint. Luckily, there is growing evidence that climate related concerns have indeed become a priced risk factor in equity returns (e.g. Bolton and Kacperczyk 2021; Hsu, Li, and Tsou 2023; Bua et al. 2024; Faccini, Matin, and Skiadopoulos 2023), putting pressure on money managers to address sustainability concerns in their investment decisions. In that regard, green equities offer investors a means to diversify conventional portfolios with assets that can potentially mitigate climate related risks in their investments (Cepni, Demirer, and Rognone 2022). These assets, however, are often exposed to not only risks associated with the prospects of the new technologies they invest in, but also the risks associated with the changing regulatory/political environment related to climate regulations. These factors, therefore, make it especially challenging to model market fluctuations in this growing segment of the equity market. Building on the recent evidence that sentiment is a significant driver of volatility and jumps in equity returns (e.g. Demirer and Prodromou 2025; Niu et al. 2024) and speculative sentiment plays a dominant role in the systematic risk of green equities (Polat, Demirer, and Ekşi 2024), this paper examines the predictive role of climate sentiment to model volatility dynamics in green equities as well as the resulting economic implications.

Utilizing two alternative measures of climate sentiment, i.e. the NYT climate and biodiversity news indexes, recently proposed by Giglio et al. (2023) based on a textual classification of positive and negative news regarding climate and biodiversity concerns, and high frequency data on green ETFs within the heterogeneous autoregressive (HAR) framework of Corsi (2008), we examine the power of climate sentiment as a predictor of volatility dynamics in green equities. Our findings show that climate and biodiversity related sentiment indeed have significant predictive power over volatility in green equity returns although these proxies can capture markedly different aspects of sentiment related to climate concerns. We find that climate sentiment that relate to risks associated with the potential negative consequences of a change in the climate system has an immediate impact on short-term volatility in green equities. While the predictive power of climate related sentiment over volatility extends to the out-of-sample volatility forecasts, we also find that the predictive power of climate sentiment can be used to extract economic gains, particularly, when it comes to bad volatility forecasts in these assets. This is an important consideration as

investors can utilize our findings to improve value at risk forecasts at short forecast horizons.

In contrast, our findings reveal that sentiment related to biodiversity loss associated with the threats to the variety of life and the ecosystem, has a significant impact on longer-term volatility in green equity returns. The long-term volatility effect of biodiversity sentiment on volatility is not unexpected considering that threats to biodiversity can have diverse economic effects ranging from medicines and drugs derived from natural compounds and the production of food and nature-based materials. We argue that as market participants respond rapidly to climate risks, investors integrate climate-related information into their trading decisions, resulting in an immediate impact on short-term volatility. However, the gradual assimilation of biodiversity sentiment into market dynamics over longer time frames involves complex regulatory, social, and technological factors that evolve gradually, leading to a slower integration into market pricing. This results in a positive impact of biodiversity sentiment on longer-term volatility.

Further analysis shows that incorporating climate and biodiversity sentiment in volatility forecasting models have economic significance, highlighted through CER and Sharpe ratios, which indicate superior performance of climate-augmented models for short and intermediate forecasts, and models integrating both sentiments for long-term strategies. Considering the growing importance of green equities in investment decisions, our findings emphasize the importance of environmental factors in financial decisions, necessitating regulatory oversight. While sentiment is found to have an impact on both good and bad volatility forecasts, our results suggest the potential amplification of volatility in green equities by positive developments in the green energy sector, contingent on market dynamics and sentiment. Furthermore, the economic significance of our findings across forecast horizons emphasizes the effectiveness of models that incorporate climate and biodiversity factors in optimizing investment performance. Future research can build on our findings to explore the predictive role of climate/biodiversity sentiment over volatility dynamics in various green equity sectors such as solar, wind, geothermal and bioenergy to devise optimal portfolio allocations to individual green sectors and examine whether these portfolios provide superior hedging benefits against climate risks. Another interesting application would be to examine whether climate/biodiversity sentiment plays

any role over risk transmissions across the green and brown equity sectors.

Notes

1. While climate risk captures the prevailing sentiment towards physical and regulatory implications of climate change on financial markets, biodiversity risk focuses on the loss of species and ecosystems rather than shifts in climate patterns.
2. For details, <https://www.invesco.com/us/financial-products/etfs/product-detail?audienceType=Investor&ticker=PBD>
3. It must be noted that we have also conducted our tests on alternative green ETFs and obtained supporting results. We provide a discussion of these findings in Section “Extension to directional volatility” under robustness checks and provide further results in an online appendix.
4. The data for the NYT climate and biodiversity news indexes is publicly available at <https://www.biodiversityrisk.org/>.
5. Our analysis follows the conventions in the volatility forecasting literature and so we have designed our sample split to ensure both robust model estimation and a meaningful evaluation of out-of-sample predictive performance.

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